BADR ALDEEN AL-KHAZAN

Operations Research & Supply Chain

Yogyakarta, Indonesia | badr.alkhazan@gmail.com | My Portfolio

SUMMARY

Final-year Industrial Engineering student (GPA 3.90/4.00), Future Global Leaders Scholarship awardee and CAPM-certified, with a research focus on Operations Research and supply chain analytics. Research on supply chain simulation, risk-aware optimization, vehicle routing, humanitarian logistics, and predictive maintenance has been accepted in a journal and at four international conferences. Fluent with AI tools to accelerate modelling, analysis and literature review.

Research interests: stochastic modelling and simulation, vehicle routing, supply chain risk and resilience, decision analytics.

EDUCATION

B.Eng. Industrial Engineering (International Program), Universitas Islam Indonesia *Expected 2027*

Scholarship: Awardee of the Future Global Leaders Scholarship (FGLS), 2023–2027.

GPA: 3.90/4.00. **Relevant coursework:** Operations Research I & II, System Simulation, System Modeling, Supply Chain System, SCM Performance Management, Data Analytics, Statistics I & II, Production Planning and Control, Research Methodology.

RESEARCH & PUBLICATIONS

Where Floods Strike and How Roads Fail: Testing Two Assumptions of Relief Pre-Positioning on the 2025 Cyclone Senyar Floods in Aceh
16th AIC 2026 (Universitas Syiah Kuala), accepted for presentation · Paper

Simulation-Based Analysis of the Bullwhip Effect in a Multi-Echelon Bottled Water Supply Chain
JTMIT, accepted, in production, 2026 · Paper

When Does Risk-Aware Workload Balancing Pay Off? A Three-City Last-Mile Routing Benchmark Using Industrial Traces
ICORSI 2026 (IIT Delhi, 21–23 Dec 2026), abstract accepted · Abstract

Safe Capacity and Time Compliance in School-Meal Kitchens: Evidence from Indonesia's Free Nutritious Meal Program
Spektrum Industri, under review, 2026 · Paper

A Configurable Electric Vehicle Routing Framework for Urban Retail Delivery with Public Charging: A Yogyakarta Case Study
In progress, 2026 · Paper

Balancing Cost and Tail Risk in the Indonesian Wheat Import Portfolio Using Nested Mean CVaR Optimization
ICOBLAST 2026 (Diponegoro University), full paper accepted · Paper

Supply Chain Governance in Sustainability Reporting: a SCOR Orchestrate Analysis of an Indonesian State-Owned Enterprise
IJPMB, under review, 2026 · Paper

Calibration and Cost-Sensitive Threshold Selection for Imbalanced Predictive Maintenance: an AI4I 2020 Study
TMIC 2026 (4th Trunodjoyo Madura International Conference), presented, Best Presenter Award · Paper

Reducing University Motorcycle Exit Time through Automated Parking Access Control (SIGAP)
JuTISI, under review, 2026 · Paper

How Depending on AI Can Impact the Analytical Skills and Critical Thinking of Young People
Published, 2026 · Paper

Germany's DIGITAL Cooperative Framework: Advancing Sustainable, Inclusive, and Innovative Global Digital Infrastructure
Published, 2025 · Paper

AWARDS & CERTIFICATIONS

- **Future Global Leaders Scholarship**: Universitas Islam Indonesia (2023–2027)
- **Outstanding Delegate**: IYEC 12, Malaysia & Singapore
- **Best Presenter**: TMIC 2026
- **Best Presentation**: EXPO RSKE 2025, Multifunctional Ergonomic Pen
- **TOEFL iBT 90/120**: ETS (2026–2028)
- **CAPM®**: Project Management Institute (2026–2029)
- **1st Place**: EXPO PSIT 2026, SIGAP RFID Parking Access
- **1st Place**: Idea Champion 2.0, AI-PHL Guardian

EXPERIENCE & LEADERSHIP

Simulation & Modelling Trainee, Delsim Laboratory, Universitas Islam Indonesia *Apr 2025 – Sep 2025*
- Built and validated discrete-event and system-dynamics models using FlexSim and PowerSim to analyze industrial system behavior.

Industrial Company Visit, PT Gaya Motor (BMW Indonesia), organized by Universitas Islam Indonesia *Apr 2025*
- Toured the assembly line and studied quality assurance systems, from parts inspection to final testing.

Deputy Head, Events Division & Industrial Engineering Ambassador, UII Global (Volunteer) *Jan 2025 – Dec 2025*
- Led academic and cultural events, logistics and partnerships; served as PIC for international-student activities.

Deputy Head, Social Responsibility Circle, Yemeni Students' Union in Indonesia (Volunteer) *Apr 2025 – Mar 2026*
- Coordinated student-support and community programs and major cultural events.

SKILLS & LANGUAGES

Methods: stochastic and mean-CVaR optimization, vehicle routing, queueing analysis, discrete-event simulation, system dynamics.
Tools: Python (OR-Tools, SciPy, pandas), FlexSim, PowerSim, Advanced Excel, generative AI tools.
Professional: Project management, team leadership, cross-cultural communication, public speaking, technical writing.
Languages: Arabic (Native), English (B2), Indonesian (B2), German (A1).